

ACTIVITY: What is the “Mystery Group”?

$*$	A	B	C	D	E	F
A	B	A	D	C	F	E
B	A	B	C	D	E	F
C	F	C	B	E	D	A
D	E	D	A	F	C	B
E	D	E	F	A	B	C
F	C	F	E	B	A	D

Definition. An isomorphism ϕ from a group G to $\bar{G}$ is:

one-to-one mapping from G onto $\bar{G}$ that is operation preserving.
 $\phi(ab) = \phi(a)\phi(b) \quad \forall a, b \in G.$

Question 1. What properties do you think isomorphic groups have?

Everything is the same!

Suppose that ϕ is an isomorphism from a group G onto a group $\bar{G}$. Then

1. ϕ carries the identity of G to the identity of $\bar{G}$.
2. For every integer n and for every group element a in G , $\phi(a^n) = [\phi(a)]^n$.
3. For any elements a and b in G , a and b commute if and only if $\phi(a)$ and $\phi(b)$ commute.
4. $G = \langle a \rangle$ if and only if $\bar{G} = \langle \phi(a) \rangle$.
5. $|a| = |\phi(a)|$ for all a in G (isomorphisms preserve orders).
6. For a fixed integer k and a fixed group element b in G , the equation $x^k = b$ has the same number of solutions in G as does the equation $x^k = \phi(b)$ in $\bar{G}$.
7. If G is finite, then G and $\bar{G}$ have exactly the same number of elements of every order.

* Are D_3 and $\mathbb{Z}_6$ isomorphic?

No!

To prove two groups are isomorphic:

Step 1: Mapping: Define $\phi: G \rightarrow \bar{G}$

Step 2: "1-1": If $\phi(a) = \phi(b)$, then $a = b$.

Step 3: onto: Let $\bar{g} \in \bar{G}$. Find $g \in G$ s.t.
 $\phi(g) = \bar{g}$. $\rightarrow$ work off to the side

Step 4: Operation Preserving: Show $\phi(ab) = \phi(a)\phi(b)$
 for all $a, b \in G$.

Example 1. Prove that $M = \left\{ \begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix} : a \in \mathbb{Z} \right\}$ under matrix multiplication is isomorphic to $\mathbb{Z}$.

- ① Define ϕ by $\phi\left(\begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix}\right) = a$. Since $a \in \mathbb{Z}$, ϕ maps from M into $\mathbb{Z}$.
- ② Let $M_1, M_2 \in M$ such that $\phi(M_1) = \phi(M_2) = b$. Then by def. of ϕ , $M_1 = \begin{bmatrix} 1 & b \\ 0 & 1 \end{bmatrix}$ and $M_2 = \begin{bmatrix} 1 & b \\ 0 & 1 \end{bmatrix}$. Thus $M_1 = M_2$ and ϕ is one-to-one.
- ③ Let $c \in \mathbb{Z}$. Observe that $\begin{bmatrix} 1 & c \\ 0 & 1 \end{bmatrix} \in M$ and $\phi\left(\begin{bmatrix} 1 & c \\ 0 & 1 \end{bmatrix}\right) = c$. Therefore, for each $c \in \mathbb{Z}$, there exists an element of M namely $\begin{bmatrix} 1 & c \\ 0 & 1 \end{bmatrix}$ such that $\phi\left(\begin{bmatrix} 1 & c \\ 0 & 1 \end{bmatrix}\right) = c$. Thus ϕ maps from M onto $\mathbb{Z}$.
- ④ Let $M_1 = \begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix}$ and $M_2 = \begin{bmatrix} 1 & b \\ 0 & 1 \end{bmatrix}$. Then:

$$\begin{aligned}
 \phi(M_1 M_2) &= \phi\left(\begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & b \\ 0 & 1 \end{bmatrix}\right) \\
 &= \phi\left(\begin{bmatrix} 1 & a+b \\ 0 & 1 \end{bmatrix}\right) \\
 &= a+b \\
 &= \phi\left(\begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix}\right) + \phi\left(\begin{bmatrix} 1 & b \\ 0 & 1 \end{bmatrix}\right) \\
 &= \phi(M_1) + \phi(M_2)
 \end{aligned}$$

Theorem (Cayley's Theorem (1854)).

Every group is isomorphic to a group of permutations.

* WHY?

Theorem 6.2 Properties of Isomorphisms Acting on Elements.

Let $G, \bar{G}$ be groups with respective identities $e, \bar{e}$. Let $k, n \in \mathbb{Z}$ and $a, b \in G$. Then

1. $\phi(e) = \bar{e}$;
2. $\phi(a^n) = [\phi(a)]^n$;
3. $ab = ba$ iff $\phi(a)\phi(b) = \phi(b)\phi(a)$;
4. $G = \langle a \rangle$ iff $\bar{G} = \langle \phi(a) \rangle$;
5. $|a| = |\phi(a)|$; and
6. $|\{x \in G \mid x^k = b\}| = |\{x \in \bar{G} \mid x^k = \phi(b)\}|$.

Exercise 1. Prove #1.

Let $\phi: G \rightarrow \bar{G}$ be an isomorphism. Then:

$\phi(e) = \phi(e \cdot e) = \phi(e)\phi(e)$. By left cancellation,

$$\bar{e} = \phi(e).$$

Exercise 2. Prove #3.

$$\phi(a^n) = \phi(a \cdot a \cdot \dots \cdot a) = \phi(a) \cdot \phi(a) \cdot \dots \cdot \phi(a) = [\phi(a)]^n.$$

Exercise 3. Prove #4.

Let $ab = ba$. Then $\phi(ab) = \phi(ba) \Rightarrow \phi(a)\phi(b) = \phi(b)\phi(a)$.

Let $\phi(a)\phi(b) = \phi(b)\phi(a)$. Then $\phi(ab) = \phi(ba)$. Since ϕ is 1-1, $ab = ba$.

Definition. An automorphism is: an isomorphism from a group G onto itself.

Exercise 4. How many automorphisms can you find from Z_{12} onto itself? What are they?

From 6.2 #4, G maps generators to generators.

$Z_{12} = \langle 1 \rangle = \langle 5 \rangle = \langle 7 \rangle = \langle 11 \rangle$. There are 4.

$$\phi(1) = 1 \quad \phi(1) = 7$$

$$\phi(1) = 5 \quad \phi(1) = 11.$$

Definition. Let G be a group and $a \in G$. The inner automorphism of G induced by a , denoted ϕ_a , is:

$$\phi_a(x) = axa^{-1}.$$

 **Exercise 5.** Determine the inner automorphism of S_3 induced by (123) .

$$S_3 = \{(1), (12), (13), (23), (123), (213)\} \quad (123)^{-1} = (213)$$

$$\phi_{(123)}(1) = (1)$$

$$\phi_{(123)}(23) = (123)(23)(213) = (13)$$

$$\phi_{(123)}(12) = (123)(12)(213) = (1)(23) = (23)$$

$$\phi_{(123)}(123) = (123)(123)(213) = (123)$$

$$\phi_{(123)}(13) = (123)(13)(213) = (12)$$

$$\phi_{(123)}(213) = (123)(213)(213) = (213)$$

Theorem.

$\text{Aut}(G)$ and $\text{Inn}(G)$ are groups under function comp.
— what are the identities? inverses?

Theorem.

For every positive integer n , $\text{Aut}(Z_n) \cong U(n)$.
— why? see above.

Additional Exercises

1. Prove that if G is a group, the mapping $\phi(g) = g^{-1}$ is an isomorphism if and only if G is Abelian.

$\Rightarrow$ Let G be a group. Suppose $\phi(g) = g^{-1}$ is an isomorphism.

Let $a, b \in G$. Then $\phi(ab) = \phi(a)\phi(b) \Rightarrow (ab)^{-1} = a^{-1}b^{-1}$.

However, by socks + shoes, $a^{-1}b^{-1} = (ba)^{-1}$. Thus

$(ab)^{-1} = (ba)^{-1} \Rightarrow ab = ba$. Thus G is Abelian.

$\Leftarrow$ Let G be Abelian. [show $\phi(g) = g^{-1}$ is an isomorphism].

① Since inverses are unique, ϕ is 1-1.

② Let $x \in G$, then $\phi(x^{-1}) = (x^{-1})^{-1} = x$. Thus G is onto itself.

③ Let $a, b \in G$. Then

$$\phi(ab) = (ab)^{-1} = b^{-1}a^{-1} \stackrel{\text{Abelian}}{=} a^{-1}b^{-1} = \phi(a)\phi(b).$$

Thus ϕ is op. pres.

Since $\phi: G \rightarrow G$ is a 1-1 + onto mapping that preserves the operation of G , ϕ is an isomorphism.

2. Determine $\text{Aut}(\mathbb{Z})$.

Generators to Generators.

$$\phi_1(1) = 1$$

$$\phi_2(1) = -1$$

$$\{\phi_1, \phi_2\}$$

3. For inner automorphism ϕ_g, ϕ_h , and ϕ_{gh} , prove that $\phi_g \phi_h = \phi_{gh}$.

Let $x \in G$. Then:

$$\phi_g[\phi_h(x)] = g(hxh^{-1})g^{-1} = (gh)x(h^{-1}g^{-1}) = (gh)x(gh)^{-1} = \phi_{gh}(x).$$

Since x is any element of G , $\phi_g(\phi_h) = \phi_{gh}$

4. Suppose $\phi \in \text{Aut}(\mathbb{Z}_n)$ and a is relatively prime to n . If $\phi(a) = b$, determine a formula for $\phi(x)$.

$$\phi(a) = \underbrace{\phi(1)\phi(1)\phi(1)\dots\phi(1)}_a = b$$

$$\Rightarrow a\phi(1) = b \Rightarrow \phi(1) = a^{-1}b$$

$$\Rightarrow \phi(x) = a^{-1}bx$$

* a^{-1} is the mult. inverse of a mod n
as $\text{U}(n)$, so this exists.

5. Let $\mathbb{R}^+$ be the positive real numbers under multiplication. Prove that $\phi(x) = \sqrt{x}$ is an automorphism of $\mathbb{R}^+$.

① Let $a, b \in \mathbb{R}^+$ such that $\phi(a) = \phi(b)$. Then
 $\sqrt{a} = \sqrt{b} \Rightarrow a = b$. Thus ϕ is 1-1.

② Let $c \in \mathbb{R}^+$. Then $c^2 \in \mathbb{R}^+$. Observe $\phi(c^2) = \sqrt{c^2} = c$.
Thus ϕ maps onto $\mathbb{R}^+$.

③ Let $a, b \in \mathbb{R}^+$. Then $\phi(ab) = \sqrt{ab} = \sqrt{a} \cdot \sqrt{b} = \phi(a)\phi(b)$.

Since ϕ is a 1-1 mapping from $\mathbb{R}^+$ onto $\mathbb{R}^+$ that is also operation preserving, ϕ is an automorphism of $\mathbb{R}^+$.

6. Suppose G is a finite Abelian group and G has no element of order 2. Show the mapping $g \rightarrow g^2$ is an automorphism of G . Is this also true of groups with infinite order? not onto

- ① let $a, b \in G$ s.t. $\phi(a) = \phi(b)$. Then $a^2 = b^2 \Rightarrow e = b^2 a^{-2} = (ba^{-1})^2$.
 Thus $|ba^{-1}| = 1$ (since $|ba^{-1}| \neq 2$ by assumption). Thus $ba^{-1} = e \Rightarrow a = b$.
 and ϕ is one-to-one.
- ② Since G is finite and ϕ is 1-1, ϕ must also be onto.
- ③ $\phi(ab) = (ab)^2 = a^2 b^2 = \phi(a)\phi(b)$. $\Rightarrow \phi$ is operation preserving.
Abelian

7. Suppose g and h induce the same inner automorphism of a group G . Prove that $h^{-1}g \in Z(G)$.

Let $x \in G$, then $g x g^{-1} = h x h^{-1} \quad \forall x \in G$.

$$\begin{aligned} g x &= h x h^{-1} g \\ h^{-1} g \cdot x &= x h^{-1} g \\ (h^{-1} g) x &= x (h^{-1} g) \end{aligned}$$

Thus $h^{-1}g$ commutes w/ all $x \in G$,
 and by definition, $h^{-1}g \in Z(G)$.

8. Prove that every automorphism of $\mathbb{R}^*$, the group of nonzero real numbers under multiplication, maps positive numbers to positive numbers and negative numbers to negative numbers.

Suppose ϕ is an automorphism of $\mathbb{R}^*$.

- ① Let $x \in \mathbb{R}^*$ and $x > 0$. Then $\phi(x) = \phi(\sqrt{x} \cdot \sqrt{x}) = \phi(\sqrt{x})\phi(\sqrt{x}) = (\phi(\sqrt{x}))^2 > 0$.
 Thus, $\phi(x)$ is positive when x is positive.
- ② Let $x \in \mathbb{R}^*$ and $x < 0$. Suppose for contradiction that $\phi(x) = a$ where $a > 0$. Then by case 1, all automorphism map a to a positive. Observe $x = \phi^{-1}(\phi(x)) = \phi^{-1}(a)$. Since ϕ^{-1} is also an automorphism on $\mathbb{R}^*$, $x = \phi^{-1}(a) > 0$, a contradiction.
 Thus $\phi(x)$ must be negative.